SAMPLE

# Waves and Sound

## Mechanical Waves, Acoustics, and Music

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Free sample: Introduction & Chapter 1
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## Introduction

Drop a stone in a still pond and the rings race outward; the water itself stays where it is. Pluck a guitar string and the air carries a message to your ear without a single molecule traveling the whole road. This is the subject of waves and sound: the traveling disturbance, the medium that hosts it, the ear that receives it, and the whole physics that connects them. This book tells the story in plain prose, the series' way: what a wave is and how one is described; the family of wave types; the speed of sound and why it changes with the medium; the history from Pythagoras's strings to Laplace's correction; Chladni's visible patterns; the ear's own remarkable mechanism; pitch and loudness and timbre; the Doppler shift; interference and beats; resonance with its famous folklore corrected; standing waves and the making of music; the sounds above and below hearing; the ocean's acoustic world; and the trivia shelf, graded honestly, the "no sound in space" folklore included. The measured facts and the honest open questions, separated from the opening page on.

The introduction's own promise being kept is worth one closing note before the walk begins. The note: the tour's carrying no equations, the four words' being the only vocabulary tax, the history's being honored, the folklore's being graded, the open questions' being labeled, and the reader's being welcome to the gentlest subject in the physics library, the sound being the science's own front porch, the porch being the tour's home ground and the walk starting on the next page, the page being the first stop and the whole subject waiting behind it, the waiting being the introduction's final promise, kept in the series' plain style from here to the goodbye.

## Chapter 1: What a Wave Is

The wave's own nature deserves the careful first statement, because the whole subject rests on one gentle paradox: the wave goes and the medium remains. The stadium crowd's doing the wave: each spectator rising and sitting in turn, the ripple racing around the ground, and no spectator having left their seat, the pattern being the traveler and the people being the stayers, the paradox being the wave's own definition: a disturbance that propagates through a medium while the medium itself merely oscillates in place. The water's rings being the surface's own local rises and falls, the ripple being the coordination, the coordination being carried from neighbor to neighbor, the neighbor-to-neighbor passing being the wave's whole mechanism, each parcel of the medium pushing the next, the push being the message, the message being the wave, and the reader's holding the paradox being the foundation: motion without travel, energy without cargo, pattern without passenger.

The reader's own second possession being the wave's two great families: the mechanical waves needing a hosting medium, the sound being the chief example, the air being the host; and the light needing none, the distinction being noted here once and honored throughout: this book being about the mechanical kind, the kind that shakes hands from parcel to parcel, the kind that cannot cross the vacuum, the kind that is the everyday world's own messenger, the families being distinguished and the tour being ready for the anatomy.

The paradox's next layer is the energy's own accounting, the cargo being the honest question. The dropped stone's push giving the water's parcels their motion, the motion's passing to the neighbors, the energy's riding the passing without any parcel making the whole trip, the shipping being the wave's own trick: the message's traveling at the wave's speed while the messengers' staying home, the energy's arrival being felt as the bobbing boat at the far end, the boat's having received a push that no water molecule carried the whole way, the delivery being the coordination's own work, the work being the wave's whole honest content, the content being thus restated: energy without traveler, effect without commuting.

The paradox's third visit is the energy's honest bookkeeping, the delivery's being the felt point. The wave's carrying power without cargo matter, the sound's carrying the speaker's energy across the room to the ear, the ear's receiving the push, the push having been passed hand to hand through the air, the chain of hands being the medium's own molecules, the molecules' never having traveled more than a whisker from home, the whisker's scale being the honest surprise: the air parcel's own displacement being microscopic while the wave's crossing the room, the disproportion being the wave's own signature, the signature being the chapter's honest third note.

The paradox's fourth visit is the reader's own felt version, the transfer being tried at home. The experiment: the slinky's lying on the floor, the one end's being pushed, the pulse's crawling to the far end, the coils' merely leaning forward and returning home, the pulse's being the only traveler, the home experiment being the foundation's own honest receipt, the receipt being felt in the hand, the hand being the chapter's own closing instrument, the instrument being the reader's own.

The paradox's fifth visit is the honest boundary note, the push's having a speed limit being the physics' own constraint. The push's traveling only as fast as the hosting medium's own message rate, the molecules' own speed being the gas's own limit, the sound's speed being close to the molecules' average pace, the closeness being no accident: the message's being carried by the messengers themselves, the messengers being the molecules, the molecules' pace being the wave's ceiling, the ceiling explaining the temperature's effect: the warmer molecules being faster, the faster messengers delivering quicker, the chain being thus complete: the medium's own physics setting the wave's own speed, the note being the foundation chapter's final honest layer.

The paradox's final visit is the wave's own honest definition being tested at the edge, the shock being the boundary case. The ordinary wave's being the gentle message, the explosion's being the violent one: the push's outrunning the ordinary message rate, the pile-up's forming the shock, the shock being the wave's own limit case, the case being mentioned in the anatomy of the extremes: the gentle ripple and the thunderclap being the same family's two ends, the ends being held together by the neighbor-to-neighbor physics, the physics being thus tested and the foundation being complete.

A further layer of the paradox is the two speeds' honest distinction, the shape's pace and the group's pace being different travelers. The wave train's envelope having its own speed, the individual crests having their own, the two being the same in the simple deep-water case and different in the dispersive ones, the dispersion being the frequency dependence of the speed, the ocean swell's spreading its long waves ahead of the short ones, the storm's distant signature being the ordered swell, the ordered arrival being the group's own arithmetic, the distinction being the advanced chapter's honest note: the wave family carrying more than one kind of speed, the more being felt in the arriving swell at the beach.

One more honest layer is the wave's own reversibility, the time-symmetric truth being the plain note. The wave's own equations not caring about time's direction, the movie's running equally well either way, the sound's being the reversible physics, the everyday irreversibility being the friction and the spreading, the spreading being the energy's own honest dilution, the distinction being the deep note told plainly: the wave's mechanics being reversible and the world's hearing being the dilution's own story, the dilution being why the shout fades and never gathers, the note being the foundation's own quietest deep layer.

The foundation's honest close is the wave's own three-question quiz, the self-test being the reader's service. The questions: does the water travel with the ripple? does the energy travel without any medium's traveling? does the speed belong to the wave or the messengers? The answers being: no, yes, and both, the three answers being the paradox's whole anatomy in pocket form, the quiz being the chapter's own final gift: the reader's testing the paradox in the pocket form, the pocket being the honest proof of ownership, the ownership being the foundation's whole honest purpose.